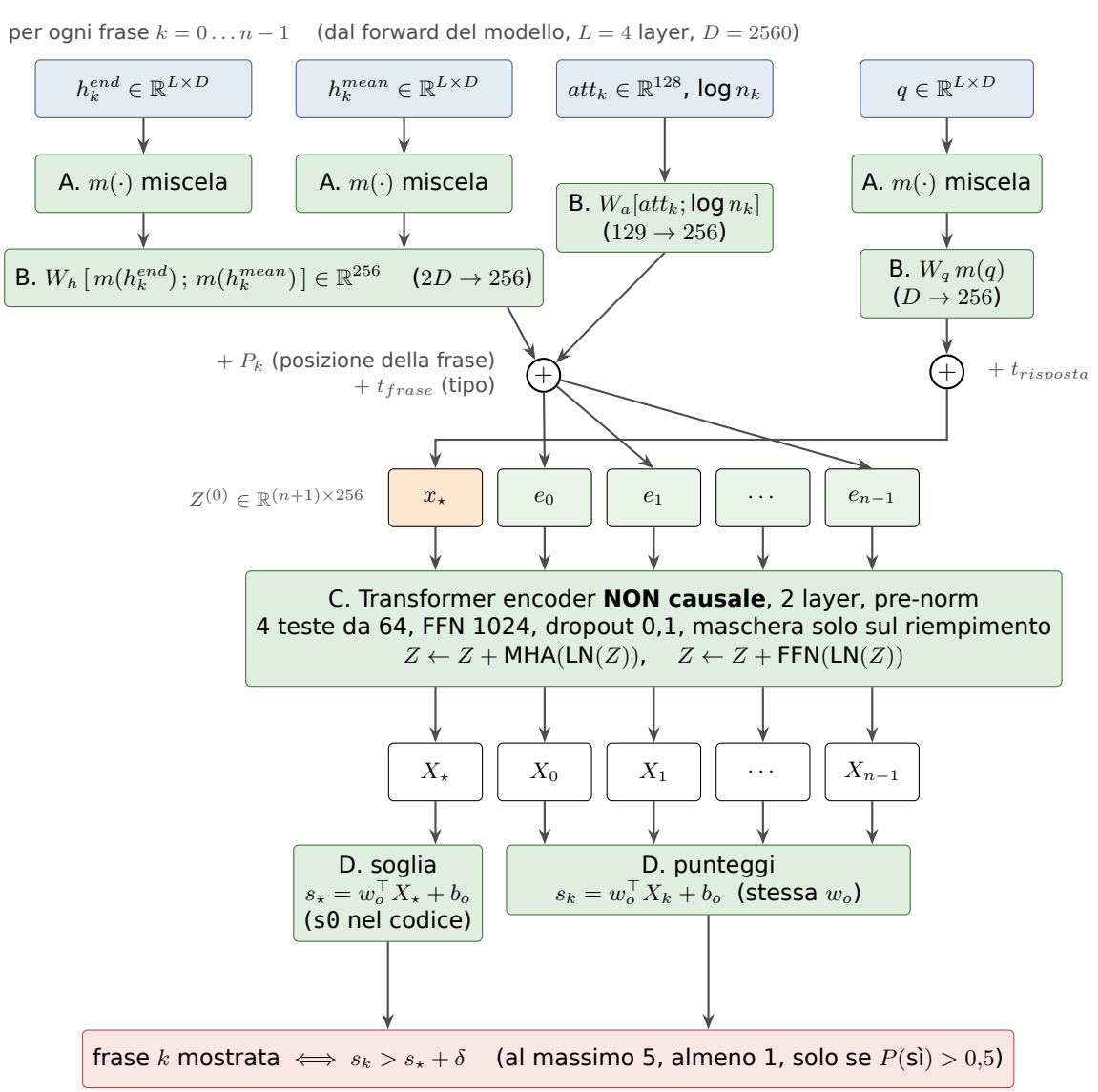
per ogni frase $k = 0 \dots n-1$ (dal forward del modello, $L = 4$ layer, $D = 2560$)
$h_k^{end} \in \mathbb{R}^{L \times D}$
$h_k^{mean} \in \mathbb{R}^{L \times D}$
$att_k \in \mathbb{R}^{128}$, $\log n_k$
$q \in \mathbb{R}^{L \times D}$
A. $m(\cdot)$ miscela
A. $m(\cdot)$ miscela
B. $W_a[att_k; \log n_k]$ $(129 \to 256)$
A. $m(\cdot)$ miscela
B. $W_h\,[\,m(h_k^{end})\,;\,m(h_k^{mean})\,] \in \mathbb{R}^{256}$ $(2D \to 256)$
B. $W_q\,m(q)$ $(D \to 256)$
$+ P_k$ (posizione della frase)
$+ t_{frase}$ (tipo)
$+ t_{risposta}$
$Z^{(0)} \in \mathbb{R}^{(n+1) \times 256}$
$x_\star$
e_0
e_1
$\cdots$
e_{n-1}
C. Transformer encoder **NON causale**, 2 layer, pre-norm
4 teste da 64, FFN 1024, dropout 0,1, maschera solo sul riempimento
$Z \leftarrow Z + \mathrm{MHA}(\mathrm{LN}(Z)), \quad Z \leftarrow Z + \mathrm{FFN}(\mathrm{LN}(Z))$
$X_\star$
X_0
X_1
$\cdots$
X_{n-1}
D. soglia
$s_\star = w_o^\top X_\star + b_o$
(s0 nel codice)
D. punteggi
$s_k = w_o^\top X_k + b_o$ (stessa w_o)
frase k mostrata $\iff$ $s_k > s_\star + \delta$ (al massimo 5, almeno 1, solo se $P(\text{sì}) > 0{,}5$)